

Assissment 1

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Branch: CSE

Semester: 3rd

Subject Name: OOPS using JAVA

UID: 24BCS11059

Section/Group: 707-SAP-B

Date of Performance: 18-02-26

Subject Code: 24CSH-207

1. Aim:

- I) Shape Area Calculation
- II) Employee Salary System 1

2. Requirements (Hardware/Software): HackerRank

3. Procedure/code:

Code I:

```
import java.io.; import java.util.;

class Shape { double area() { return 0; } }

class Rectangle extends Shape { double length, breadth;

Rectangle(double length, double breadth) {
    this.length = length;
    this.breadth = breadth;
}
@Override
double area() {
    return length * breadth;
}

}

class Circle extends Shape { double radius;

Circle(double radius) {
    this.radius = radius;
}

@Override
```

```

double area() {
    return Math.PI * radius * radius;
}
}

```

```

public class Solution { public static void main(String[] args) { Scanner sc =
new Scanner(System.in);

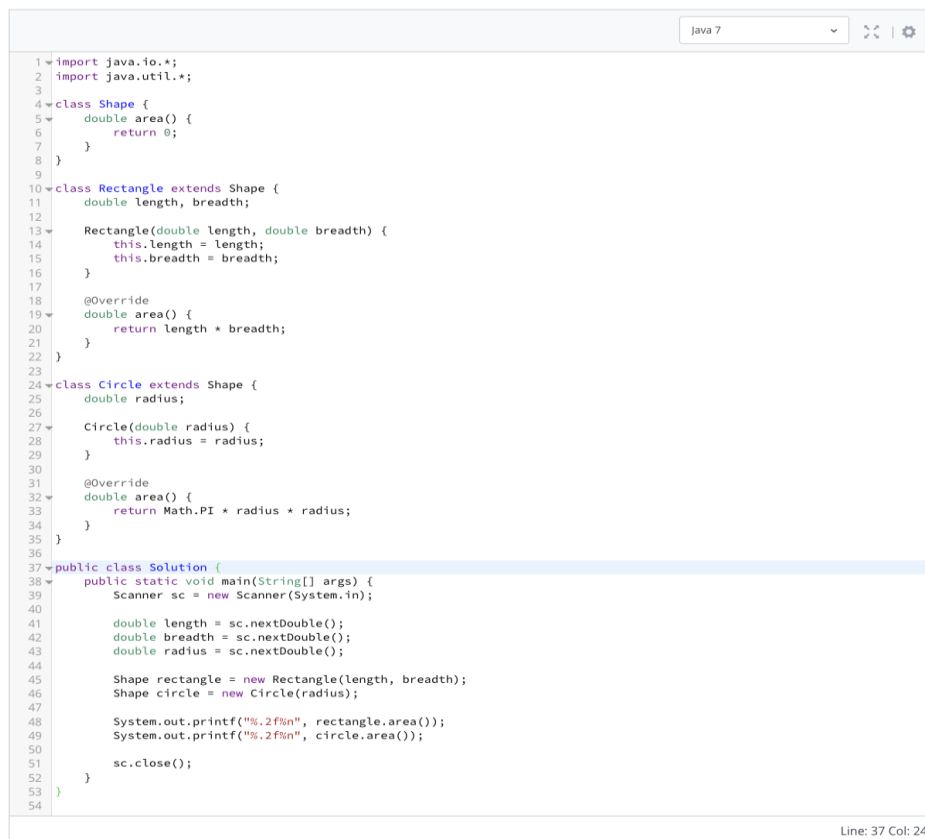
```

```

    double length = sc.nextDouble();
    double breadth = sc.nextDouble();
    double radius = sc.nextDouble();
    Shape rectangle = new Rectangle(length, breadth);
    Shape circle = new Circle(radius);
    System.out.printf("%.2f\n", rectangle.area());
    System.out.printf("%.2f\n", circle.area());
    sc.close();
}
}

```

Screenshot:



```

1  import java.io.*;
2  import java.util.*;
3
4  class Shape {
5      double area() {
6          return 0;
7      }
8  }
9
10 class Rectangle extends Shape {
11     double length, breadth;
12
13     Rectangle(double length, double breadth) {
14         this.length = length;
15         this.breadth = breadth;
16     }
17
18     @Override
19     double area() {
20         return length * breadth;
21     }
22 }
23
24 class Circle extends Shape {
25     double radius;
26
27     Circle(double radius) {
28         this.radius = radius;
29     }
30
31     @Override
32     double area() {
33         return Math.PI * radius * radius;
34     }
35 }
36
37 public class Solution {
38     public static void main(String[] args) {
39         Scanner sc = new Scanner(System.in);
40
41         double length = sc.nextDouble();
42         double breadth = sc.nextDouble();
43         double radius = sc.nextDouble();
44
45         Shape rectangle = new Rectangle(length, breadth);
46         Shape circle = new Circle(radius);
47
48         System.out.printf("%.2f\n", rectangle.area());
49         System.out.printf("%.2f\n", circle.area());
50
51         sc.close();
52     }
53 }
54

```

Line: 37 Col: 24

Code II:

```
import java.io.*;
import java.util.*;
```

```
class Employee {
    double basicSalary;
```

```
Employee(double basicSalary) {
    this.basicSalary = basicSalary;
}
```

```
double calculateSalary() {
    return basicSalary;
}
}
```

```
class Manager extends Employee {
    double bonus;
```

```
Manager(double basicSalary, double bonus) {
    super(basicSalary);
    this.bonus = bonus;
}
```

```
@Override
double calculateSalary() {
    return basicSalary + bonus;
}
}
```

```
public class Solution {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        double basicSalary = sc.nextDouble();
```

```
double bonus = sc.nextDouble();
```

```
Employee manager = new Manager(basicSalary, bonus);
```

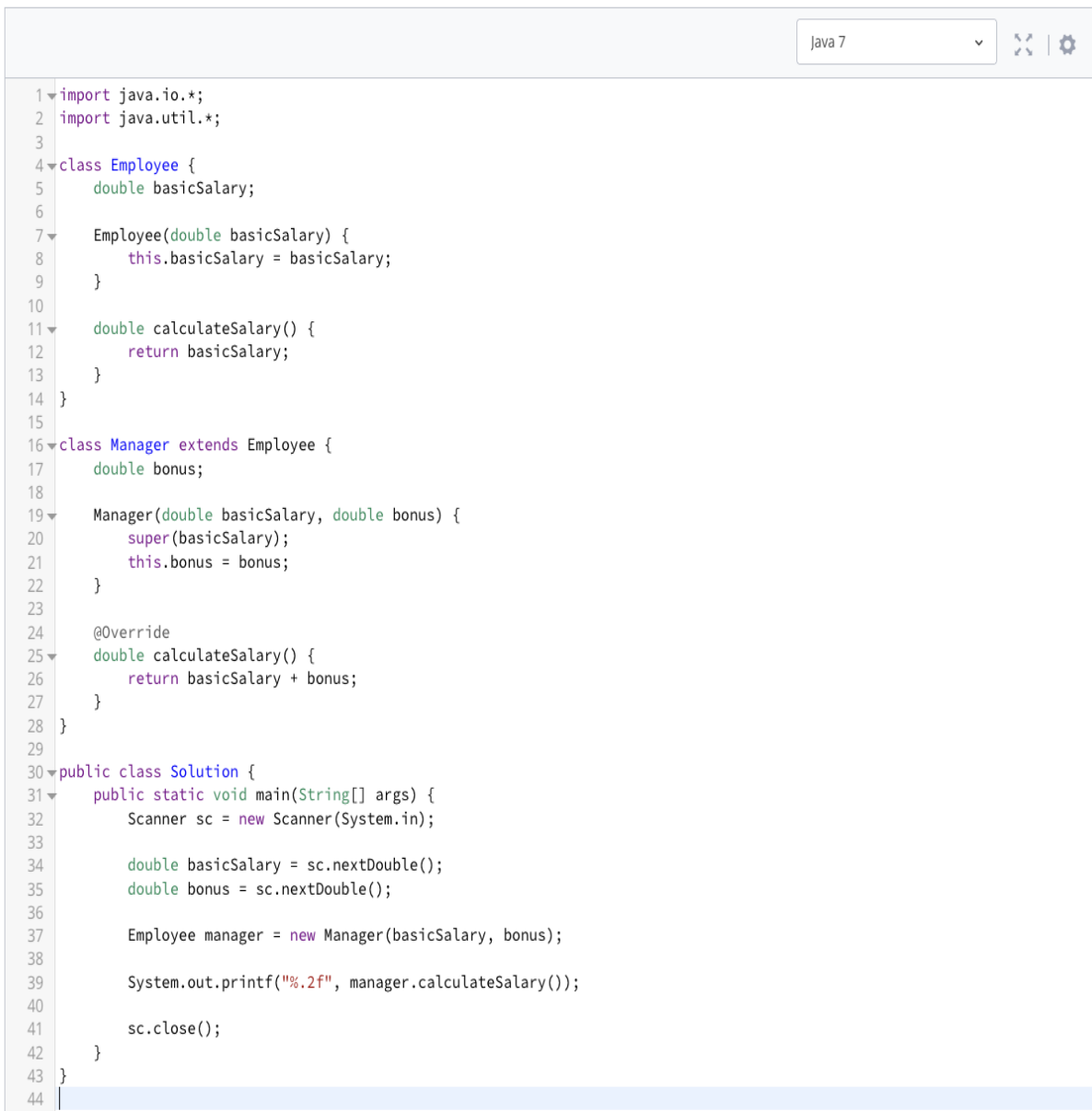
```
System.out.printf("%.2f", manager.calculateSalary());
```

```
sc.close();
```

```
}
```

```
}
```

Screenshot:



```
1 import java.io.*;
2 import java.util.*;
3
4 class Employee {
5     double basicSalary;
6
7     Employee(double basicSalary) {
8         this.basicSalary = basicSalary;
9     }
10
11     double calculateSalary() {
12         return basicSalary;
13     }
14 }
15
16 class Manager extends Employee {
17     double bonus;
18
19     Manager(double basicSalary, double bonus) {
20         super(basicSalary);
21         this.bonus = bonus;
22     }
23
24     @Override
25     double calculateSalary() {
26         return basicSalary + bonus;
27     }
28 }
29
30 public class Solution {
31     public static void main(String[] args) {
32         Scanner sc = new Scanner(System.in);
33
34         double basicSalary = sc.nextDouble();
35         double bonus = sc.nextDouble();
36
37         Employee manager = new Manager(basicSalary, bonus);
38
39         System.out.printf("%.2f", manager.calculateSalary());
40
41         sc.close();
42     }
43 }
44
```

4. Output:

I:

Line: 1, Col: 1

Test against custom input

Run Code

Submit Code

Upload Code as File

50000
10000

Custom Testcase

Compilation Successful

Input (stdin)
50000
10000

Your Output
60000.00

II:

Line: 1, Col: 1

Test against custom input

Run Code

Submit Code

Upload Code as File

5 10
7

Custom Testcase

Compilation Successful

Input (stdin)
5 10
7

Your Output
50.00
153.94

5. Learning Outcome:

- Understand **inheritance** by extending base classes (Employee, Shape) into specialized subclasses (Manager, Rectangle, Circle).
- Learn **method overriding** to provide class-specific implementations (calculateSalary(), area()).
- Apply **runtime polymorphism**, where a superclass reference calls subclass methods.
- Practice **constructor chaining** using super to initialize parent class data.
- Gain experience with **object-oriented design** and clean input/output handling in Java.

